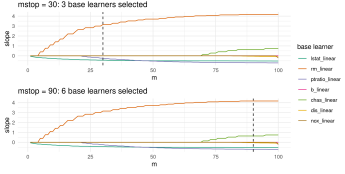
Introduction to Debugging Lectures

Demo Lecture
Layout Macro Testing With Cannibalized Content



Learning goals

- Test layout
- Fix stuff
- Go home

DEMO SLIDES

This is a demo lecture chunk.
It is recommended to view these slides as PDF and LaTeX code side by side
Refer to the `slds-lmu/lecture_debug` repository for the source and PDF:

- Compiled PDF
- LaTeX source

\SPLITV(TCB)(TCB)

- `\splitVCC` creates two centered columns
- `\splitVTT` creates two top-aligned columns
- `\splitVBB` for bottom-aligned columns

Left column these two columns should both be top-aligned with their unequal content lengths

- Example itemize content for centered columns
- Second itemize item

Right column Lorem ipsum dolor sit amet, consectetur adipiscing elit

Lorem ipsum dolor sit amet, consectetur adipiscing elit

\SPLITV WITH UNEQUAL SIZES

You only need to specify the width of the first column:

`\splitVTT[0.75]{left content}{right content}` creates a column with 75% width and the second column will fill the remaining space:

First column with 75% of the text width on the slide

Second column with rest

Tiny column

Second column with a lot of room for activities

TWO COLUMNS, MINIMAL ADJUSTMENTS

Compact version if you do not want to use the full slide width: `\splitVCompact{0.2}{0.2}` only takes up 20% of the slide width in each column for a total of 40% with a minimal horizontal spacer:

First column with 20%

Second column with 20%

3 COLUMNS

Using `\splitVThree{Left}{Middle}{Right}` for 3 equally sized columns:

First column content is here

Second column content is here as well

And also a third column is here just in case

Then there is `\splitVThreeCustom` for three columns of arbitrary widths, where the width of the third column can also be inferred from the first two

first column with 20% width

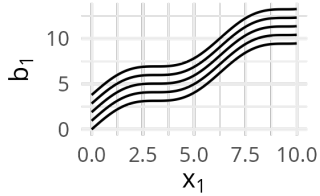
second column with 20% width

third column with remaining width

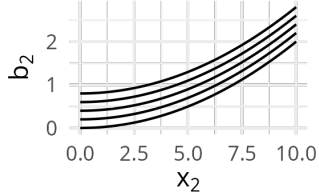
TWOBYTWO 2×2 LAYOUT / QUADRANTS

Use `\twobytwo{top left}{top right}{bottom left}{bottom right}` for horizontally aligned quadrants

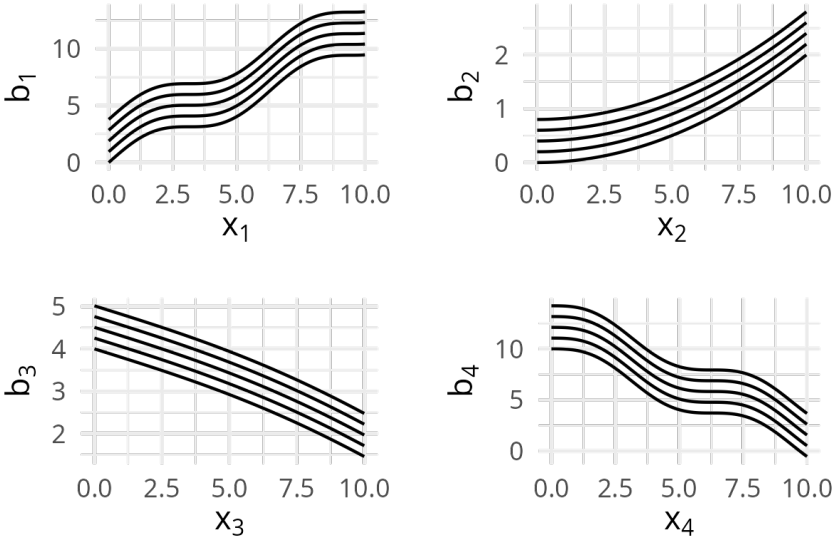
- Example top left...
- ...quadrant content
- Next to a figure

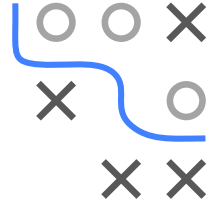


- Bottom left...
- ...quadrant content



2×2 LAYOUT: ALL IMAGES





ITEMIZE WRAPPERS

Presets fro `itemize` with different vertical spacings (`itemsep`). The default value for `itemsep` is apparently `2pt`. Use `\the \itemsep` to find out the current value.

- **Default itemize**
- itemsep is unmodified
- Another thing
- Words
- A fourth thing to show
- Just filling space here
- Hello there

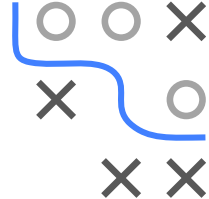
- **Uses environment** `itemizeS`
- itemsep is: -2.0pt
- Another thing
- Words
- A fourth thing to show
- Just filling space here
- Hello there

ITEMIZEM AND ITEMIZEL

- **Default itemize** with `itemizeM`
- itemsep is unmodified
- Another thing
- Words
- A fourth thing to show
- Just filling space here
- Hello there

- **Uses environment** `itemizeL`
- itemsep is: 15.0pt
- Another thing
- Words
- A fourth thing to show
- Just filling space here
- Hello there

ITEMIZEF FOR VERTICAL FILLING



- **Uses environment** `itemizeF`
- itemsep is: 10.07397pt plus 1.0fill
- Automatically uses all vertical space
- Depends on containing enviornment though!

ITEMIZE SPACING AND FONT SIZE CONTROL

itemizeS with small text

- Small font + compact spacing
- Use for detailed lists
- When every line matters

itemizeL with large text:

- Large font + wide spacing
- Makes content stand out
- Ideal for takeaways

itemizeM with default text:

- Default sizing and spacing works best for most content
- Provides good readability and balanced slide appearance

itemizeM with footnotesize text:

- Default sizing and spacing works best for most content
- Provides good readability and balanced slide appearance

REFERENCE BUTTONS ▸ Author 2025

- Use `\furtherreading` with citation key:
  - Example: ▸ Author 2025
  - Basically same as old citelink / citebutton
- Use `\sourceref` with URL:
  - Direct URL: ▸ Click for source
  - URL with www: ▸ Click for source
- Or use `\sourceref` with citation key:
  - Example: ▸ Click for source
  - Another example: ▸ Click for source
  - Uses the url field in bib file for link:

The buttons have different background colors:

- Light blue background for further reading
- Light orange/peach background for source references

IMAGE TAKING THE FULL SLIDE WIDTH

IMAGE USING HALF THE AVAILABLE WIDTH

CENTERED IMAGE USING HALF THE AVAILABLE WIDTH

FULL WIDTH CENTERED IMAGE WITH CITEKEY ATTRIBUTION

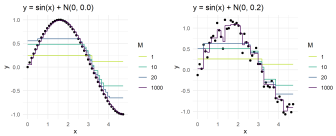
▸ Click for source

IMAGES ATTRIBUTION CAN BE URL OR CITEKEY

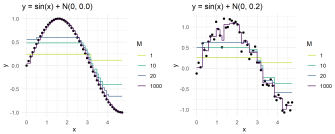
▸ Click for source

▸ Click for source

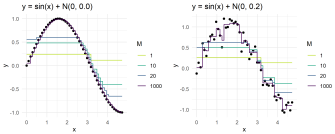
IMAGES WITH ALIGNMENT AND ATTRIBUTION



Click for source



Click for source



Click for source

IMAGEFIXED: ABSOLUTE POSITIONING

Use `\imageFixed{x}{y}[width][attribution]{file}` for absolute positioning:

- `x` and `y` are absolute coordinates from top-left
- Coordinates can be lengths like `2cm` or relative like `0.1\paperwidth`
- Width is optional (default 1.0 relative to `\textwidth`)
- Attribution works the same as other image macros
- Useful for overlaying images at precise positions

IMAGEFIXED EXAMPLES

Top-left corner:

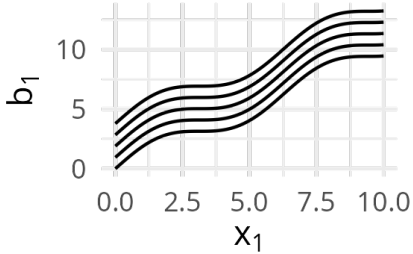
Center-right using paper dimensions:

Bottom-left with attribution:

▸ Click for source

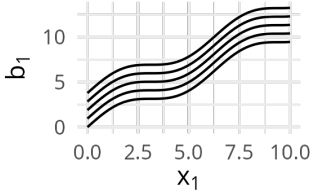
IMAGES WITHIN ITEMIZE

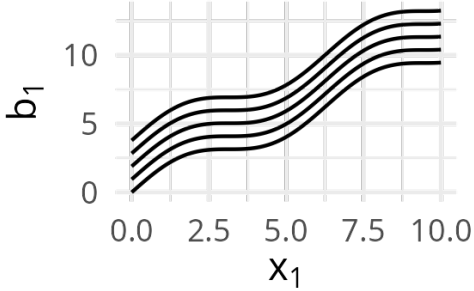
- Foo bar
- Plubber

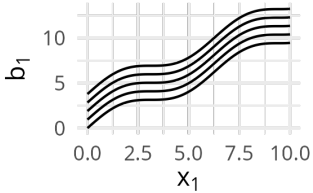


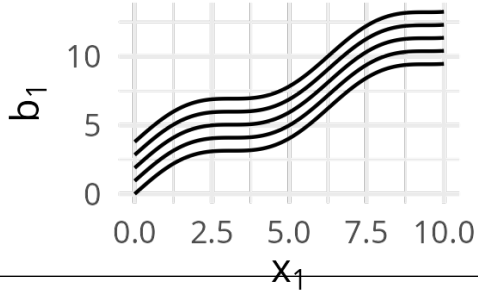
- jfowiehfgnsdlkjnfg

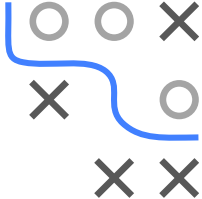
IMAGES WITHIN SPLITV











IMAGES WITHIN SPLITVCC

▸ Click for source

FRAMEI ENVIRONMENT

- This is a frame consisting only of an itemize environment
- Nothing else here, just itemize.
- Default center alignment

- Look, an image

FRAMEI ENVIRONMENT

- This is a frame consisting only of an itemize environment
- nested items
- Look, an image

FRAMEI ENVIRONMENT

- This is a frame consisting only of an itemize environment

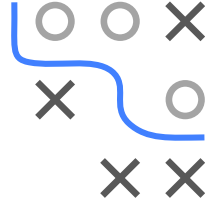
  - nested items

- Look, an image

CUSTOMIZED FRAMEI ENVIRONMENT

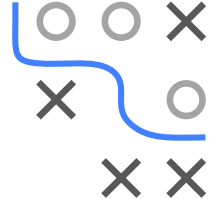
- The `framei` environment supports key-value arguments
- Font size: `fs=small` (or footnotesize, large, etc.)
- Separation: `sep=S` (or M, L, F)
- This example uses `framei[fs=small,sep=S]`
- Perfect for overview slides or detailed lists
- Makes it easy to fit more content on a single slide
- Please don't overdo it

ANOTHER FRAMEI EXAMPLE



- Larger text with wider spacing
- Passed directly to the `framei` environment
- Using `[fs=large,sep=L]`
- For emphasis and impact

A FILLING FRAMEI EXAMPLE

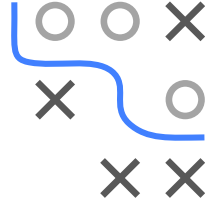


- Larger text with widest spacing
- Passed directly to the `framei` environment
- Using `[fs=large,sep=F]`
- For the glory of hypnotoad

FRAMEI WITH TOP ALIGNMENT

- The `align=top` option aligns content to the top of the frame
- Useful for consistency when not filling the entire frame
- Pairs well with `fs` and `sep` options
- Example: `framei[align=top]`

FRAMEI WITH BOTTOM ALIGNMENT



- Bottom alignment with `align=bottom`
- Positions content at the bottom of the frame
- Can be combined with other options

FRAMEI: COMBINED OPTIONS

- This demonstrates combining multiple options
- Using `[fs=small,sep=S,align=top]`
- Small font size for more content
- Compact spacing (S)
- Top-aligned positioning
- Perfect for comprehensive slides

INTEGRATION WITH FRAMEI ENVIRONMENT

- The framei environment creates an itemize list
- We can nest splitV inside list items
- Using splitVTT within a list item:

▸ Click for source

- This continues the itemize list
- No need for a nested itemize
- The items appear properly beside the image

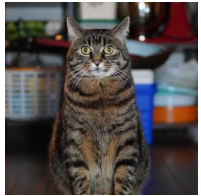
- Back to the main list
- With proper spacing between elements

FRAMEI WITH NESTED ITEMIZE AND FONT SIZE CONTROL

- Item A
  - itemize inherits footnotesize
- Global item
  - itemizeM overrides with tiny fontsize
- Another global item
  - itemizeM inherits footnotesize
- Item B

SPLITV INSIDE REGULAR ITEMIZE

- First regular bullet point
- Some more text in an item



- This appears beside the image
- No need for a nested itemize environment

- Assume data is separable
- We can continue the list afterward
- Another example with top alignment:

- Top-aligned content
- In a normal itemize environment
- Final regular bullet point

FRAMEV LIKE FRAME BUT WITH FONT SIZE PARAMETER

This entire frame uses the small font size.

- this regular itemize list uses the small font size
- you can nest things in here as needed

- this itemizeM list overrides the font size
- if you use itemizeM etc. they just inehrit the font size as well

FRAMEV WITH NESTED REGULAR ITEMIZES

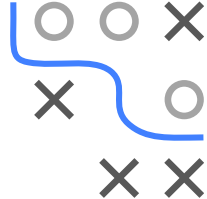
- Item A
  - Subitem 1
- Item B

FRAMEV WITH TOP ALIGNMENT

This framev uses `[align=top]` to position content at the top of the frame.
Unlike framei, framev does not automatically wrap content in an itemize environment.

- You can add itemize lists manually
- Top alignment applies to the entire frame content
- Combine with font size: `[fs=small,align=top]`

FRAMEV WITH BOTTOM ALIGNMENT



Content positioned at the bottom using `[align=bottom]`.
This is useful for emphasis or visual variety in presentations.

FRAMEV: SMALL FONT + TOP ALIGNMENT

This demonstrates combining font size and alignment options: `[fs=small,align=top]`

- Small font size makes room for more content
- Top alignment keeps content consistently positioned
- framev provides flexibility without automatic itemize wrapper

You can also use shorthand: `[align=t]` instead of `[align=top]`
Similarly: `align=c` for center, `align=b` for bottom

VERBATIM CONTENT

- Verbatim does not work with framei or framev
- But you can mix itemizeL etc and verbatim
- The important thing is to use "normal" frame environment with `fragile,c`

```
#include<stdio.h>

int main() {
 printf("Hello World\n");
 return 0;
}
```